

LIMITS OF PROOFS: SOCIAL FOUNDATIONS OF CRYPTOGRAPHY

ADVANCED TOPICS IN ~~CYBERSECURITY~~ CRYPTOGRAPHY (7CCSMATC)

Martin R. Albrecht

The Indictment

Example: Non-Repudiation

Example: At-Compromise Security

THE INDICTMENT

DIGITAL SIGNATURES: WHAT SHOULD THEY DO?

What do you think should be the security goal of a digital signature scheme?

New Directions in Cryptography

Invited Paper

WHITFIELD DIFFIE AND MARTIN E. HELLMAN, MEMBER, IEEE

Abstract—Two kinds of contemporary developments in cryptography are examined. Widening applications of teleprocessing have given rise to a need for new types of cryptographic systems, which minimize the need for secure key distribution channels and supply the equivalent of a written signature. This paper suggests ways to solve these currently open problems. It also discusses how the theories of communication and computation are beginning to provide the tools to solve cryptographic problems of long standing.

The best known cryptographic problem is that of privacy: preventing the unauthorized extraction of information from communications over an insecure channel. In order to use cryptography to insure privacy, however, it is currently necessary for the communicating parties to share a key which is known to no one else. This is done by sending the key in advance over some secure channel such as private courier or registered mail. A private conversation between two people with no prior acquaintance is a common occurrence in business, however, and it is unrealistic

“THE EQUIVALENT OF A WRITTEN SIGNATURE” [DH76]

*We note that the formulation of digital signatures also provides a clear statement of the **essential ingredients of handwritten signatures**. The ingredients are each person’s ability to sign for him/herself, a universally agreed-upon verification procedure, and the belief (or assertion) that it is infeasible (or at least hard) to forge signatures in a manner that passes the verification procedure. It is **not clear to what extent handwritten signatures do meet these requirements**. In contrast, our treatment of digital-signature schemes provide precise statements concerning the extent to which digital signatures meet these requirements.¹*

¹Oded Goldreich. **Foundations of Cryptography: Basic Applications**. Vol. 2. Cambridge, UK: Cambridge University Press, 2004. ISBN: 978-0511721656. DOI: 10.1017/CB09780511721656, p.498, emphasis added.

MAIN REFERENCE



Jean-François Blanchette. **Burdens of proof: Cryptographic culture and evidence law in the age of electronic documents.** MIT Press, 2012, Chapter 4: The Equivalent of a Written Signature

“ON THESE QUESTIONS, THE LITERATURE REMAINS SILENT.”

Goldreich has argued cryptographic practice proceeds through “the identification, conceptualization and rigorous definition of cryptographic tasks which capture natural security concerns.” [...] How does one identify such “natural security concerns”? On these questions, the literature remains silent. Furthermore, neither of these approaches provide a rationale for the definitions of the “mutant” signatures described in the preceding section, which can hardly be accounted for on the basis of “natural requirements.” — [Bla12, p.89]

“‘MUTANT’ SIGNATURES”

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SOME ADVANCED SIGNATURE NOTIONS

Group ... allow any member of the group to sign a document, but it is not possible to identify who did. If a dispute arises over the legitimacy of a signature, a group manager (who controls membership) can be called on to identify the signer.

Ring ... require no group managers and no prespecified group size.

Blind ... allow the signer affixes her signature to a document but they cannot later link the two together.

Designated-Verifier ... allow a receiver to convince themselves that a message is not a forgery, but cannot transfer that conviction to a third party.

These signature schemes represent fascinating new configurations of responsibility, liability, trust, and power within the signing process. In most cases, there are no obvious “real-world” equivalents to these mathematical constructs, and in most cases, it is difficult to imagine the specific context in which they might be applied. Nevertheless, it is standard practice for cryptographic papers to justify such schemes with a “motivation” narrative, a “real-world” scenario that aims to suggest a plausible practical application of the signature scheme. — [Bla12, p.82]

To motivate the title for this paper, suppose that Bob (also known as “Deep Throat”) is a member of the cabinet of Lower Kryptonite, and that Bob wishes to leak a juicy fact to a journalist about the escapades of the Prime Minister, in such a way that Bob remains anonymous, yet such that the journalist is convinced that the leak was indeed from a cabinet member.

[...]

A standard group signature scheme does not solve the problem, since it requires the prior cooperation of the other group members to set up, and leaves Bob vulnerable to later identification by the group manager, who may be controlled by the Prime Minister.

[...]

The correct approach is for Bob to send the story to the journalist (through an anonymizer), signed with a ring signature scheme that names each cabinet member (including himself) as a ring member. The journalist can verify the ring signature on the message, and learn that it definitely came from a cabinet member. He can even post the ring signature in his paper or web page, to prove to his readers that the juicy story came from a reputable source. However, neither he nor his readers can determine the actual source of the leak, and thus the whistleblower has perfect protection even if the journalist is later forced by a judge to reveal his “source” (the signed document).

THE CRYPTOGRAPHIC PAPER GENRE

Like so many modular Lego pieces, cryptographic primitives and design patterns are assembled in new schemes and protocols exhibiting security properties with no obvious real-world equivalents. This creative process is one of the core professional activities of cryptographers, rewarded through conference presentations, journal publications, and commercial patents. Yet the cryptographic paper genre seems to require that these products of mathematical creativity be justified in some “real-world” setting, motivated either by their potential application, their evidential value, or the new threats they identify. These justificatory scenarios are remarkable in their assumptions that the properties of cryptographic objects, as designed and discussed by cryptographers, will translate transparently into the complex social settings they describe. — [Bla12, p.84]

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A CAVEAT

The claim is true for many but not all cryptographic papers, e.g. [GGSW13] starts with:

When we encrypt a message using a public-key encryption scheme, we allow the receiver to learn our message only if he knows a secret key corresponding to his public key. What if we don't really care if he knows a secret key, but we do care if he knows a solution to a crossword puzzle that we saw in the Times? Or if he knows a short proof for the Goldbach conjecture? Or, in general, the solution to some NP search problem? In this paper, we ask the question: Can we encrypt a message so that it can only be opened by a recipient who knows a witness to an NP relation?

THE INDICTMENT

- The field of cryptography prides itself on its rigour, rightly so, as hopefully demonstrated in this module.
- Blanchette points out that this rigour is sorely lacking when it comes to the social relations that cryptography models.

It is our opinion that the design of cryptographic systems has to be based on firm foundations; whereas ad-hoc approaches and heuristics are a very dangerous way to go.^a

^aOded Goldreich. **On the Foundations of Modern Cryptography (Invited Lecture)**. In: CRYPTO'97. Ed. by Burton S. Kaliski Jr. Vol. 1294. LNCS. Springer, Berlin, Heidelberg, Aug. 1997, pp. 46–74. DOI: 10.1007/BFb0052227.

Interim Conclusion

Cryptography is also a social science unaware of itself.

TWO DELINEATIONS

Studying the social foundations of cryptography with same rigour as we study the complexity-theoretic or mathematical foundations of cryptography is not an area of ...

- **human-computer interaction** (HCI) such as usable security.
- There is no computer to interact with when we ask “what is a digital signature scheme?”
- **applied cryptography**, which applies cryptographic solutions to (presumed-)real settings.
- Sciences tend to have an “applied” department, but while Alice and Bob do not live in physics or mathematics paper introductions, they do frequently appear in cryptography.
- The definition of a ring signature and its security properties is a definitional task, not an application of cryptography.

EXAMPLE: NON-REPUDIATION

DISPUTES BETWEEN TRANSMITTER AND RECEIVER

Authentication is at the heart of any system involving contracts and billing. Without it, business cannot function. Current electronic authentication systems cannot meet the need for a purely digital, unforgeable, message dependent signature. They provide protection against third party forgeries, but do not protect against disputes between transmitter and receiver. — [DH76]

If electronic mail systems are to replace the existing paper mail system for business transactions, “signing” an electronic message must be possible. The recipient of a signed message has proof that the message originated from the sender. This quality is stronger than mere authentication [...]; the recipient can convince a “judge” that the signer sent the message. To do so, he must convince the judge that he did not forge the signed message himself! — [RSA78]

NON-REPUDIATION

- [DH76] and [RSA78] were both concerned with what is called “non-repudiation” in law.
- They envisioned contracts and the disputes arising from them as a major area of contention where cryptography would regulate behaviour.
- Their vision did not pan out:
 1. The standard notion of security for a signature scheme is EUF-CMA and it is not immediately clear how it captures non-repudiation, at least under the law.
 2. E-commerce took off without cryptographic digital signatures under contracts being common.

4.B.2 Security Definition for Digital Signatures NIST intends to standardize one or more schemes that enable existentially unforgeable digital signatures with respect to an adaptive chosen message attack. (This property is generally denoted *EUFCMA security* in academic literature.)

The above security definition should be taken as a statement of what NIST will consider to be a relevant attack. Submitted algorithms for digital signatures will be evaluated based on how well they appear to provide this property when used as specified by the submitter. Submitters are not required to provide a proof of security, although such proofs will be considered if they are available.

For the purpose of estimating security strengths, it may be assumed that the attacker has access to signatures for no more than 2^{64} chosen messages; however, attacks involving more messages may also be considered. Additionally, it should be noted that NIST is primarily concerned with attacks that use classical (rather than quantum) queries to the signing oracle.

RECAP: EXISTENTIAL UNFORGEABILITY UNDER CHOSEN MESSAGE ATTACKS (EUF-CMA)

EUF-CMA	$S(m)$
1: $\mathcal{Q} \leftarrow \emptyset$	1: $\sigma \leftarrow \Sigma.\text{Sign}(\text{sk}, m)$
2: $\text{vk}, \text{sk} \leftarrow \Sigma.\text{KeyGen}(1^\lambda)$	2: $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{(m, \sigma)\}$
3: $(m^*, \sigma^*) \leftarrow \mathcal{A}^S(\text{vk})$	3: return σ
4: $b \leftarrow \Sigma.\text{Verify}(\text{vk}, m^*, \sigma^*)$	
5: return $b = 1 \wedge (m^*, \cdot) \notin \mathcal{Q}$	

$$\text{Adv}_{\mathcal{A}, \Sigma}^{\text{euf-cma}}(\lambda) := \Pr[\text{EUF-CMA}_{\Sigma}^{\mathcal{A}}(\lambda) \Rightarrow 1]$$

STRONG DESTRUCTIVE EXCLUSIVE OWNERSHIP (S-DEO)

S-DEO	$S(m)$
1: $\mathcal{Q} \leftarrow \emptyset;$	1: $\sigma \leftarrow \Sigma.\text{Sign}(\text{sk}, m)$
2: $\text{vk}, \text{sk} \leftarrow \Sigma.\text{KeyGen}(1^\lambda);$	2: $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{(m, \sigma)\}$
3: $(m^*, \sigma^*, \text{vk}^*) \leftarrow \mathcal{A}^S(\text{vk});$	3: return σ
4: $b \leftarrow \Sigma.\text{Verify}(\text{vk}^*, m^*, \sigma^*)$	
5: return $b \wedge (\exists m \neq m^* : (m, \sigma^*) \in \mathcal{Q}) \wedge \text{vk}^* \neq \text{vk}$	

$$\text{Adv}_{\mathcal{A}, \Sigma}^{\text{s-deo}}(\lambda) := \Pr[\text{S-DEO}_{\Sigma}^{\mathcal{A}}(\lambda) \Rightarrow 1]$$

EUFCMA $\not\Rightarrow$ S-DEO

- We have seen (in the “rewinding” lecture) that Schnorr signatures are EUFCMA in the ROM and if discrete logarithms are hard.
- Below we give an attack breaking S-DEO.

RECAP: SCHNORR

Let $H : \mathbb{G} \times \{0, 1\}^* \rightarrow \mathbb{Z}_p$ be a hash function

Gen

$sk := x \leftarrow \$ \mathbb{Z}_p; \quad vk := X := G^x$

Claus-Peter Schnorr. **Efficient Identification and Signatures for Smart Cards**. In: *CRYPTO'89*. Ed. by Gilles Brassard. Vol. 435. LNCS. Springer, New York, Aug. 1990, pp. 239–252. DOI: 10.1007/0-387-34805-0_22

Sign(sk, m)

1. $y \leftarrow \$ \mathbb{Z}_p$ and set $Y \leftarrow G^y$
2. $c \leftarrow H(Y, m)$
3. $z \leftarrow y - c \cdot x$

$\sigma := (Y, z)$

Verify(vk, m, σ)

1. $c \leftarrow H(Y, m)$
2. $Y \stackrel{?}{=} G^z \cdot X^c = G^z \cdot G^{c \cdot x} = G^{y - c \cdot x + c \cdot x}$

S-DEO ATTACK

Attack:

1. Pick m, m^* .
2. Call $S(m)$ to get $\sigma^* := (Y := G^y, z)$.
3. $c^* \leftarrow H(Y, m^*)$
4. Set $vk^* := vk^{c/c^*}$
5. Output (m^*, σ^*, vk^*)

Verification:

$$\begin{aligned} c^* &\leftarrow H(Y, m^*) \\ Y &= G^z \cdot (vk^*)^{c^*} \\ &= G^z \cdot G^{c^* \cdot (c/c^*) \cdot x} \\ &= G^z \cdot G^{c \cdot x} \\ &\stackrel{\checkmark}{=} G^{y - c \cdot x + c \cdot x} \end{aligned}$$

Easy Fix [CDFFJ21]

$$c \leftarrow H(Y, m, X)$$

The relationship between S-DEO and non-repudiation is, for example, discussed in Neal Koblitz and Alfred Menezes. **Another Look at Security Definitions**. Cryptology ePrint Archive, Report 2011/343. 2011. URL: <https://eprint.iacr.org/2011/343>²

²Published as Neal Koblitz and Alfred Menezes. **Another look at security definitions**. In: *Adv. Math. Commun.* 7.1 (2013), pp. 1–38. DOI: 10.3934/AMC.2013.7.1. URL: <https://doi.org/10.3934/amc.2013.7.1>.

we have to worry about another damaging potential use of [attacks on S-DEO]: to undermine non-repudiation.

- Bob signs a contract m with vk and later wants to claim he did not sign it.
- Bob uses the S-DEO attack to generate another public key and message for the same signature σ^* .
- Bob argues in court that there is another person holding vk^* who could have made that signature.
- This strategy might not work since a genuine σ^* must have been seen by whoever produced vk^* , otherwise EUF-CMA would not hold.

If the lawyer says that perhaps someone was trying to harm Bob by falsely transmitting a promise that appeared to come from him, then Alice's lawyer might get the expert witness under cross-examination to admit that a duplicate signature is possible only if Bob's valid signature was first available to the attacker. [...] So if Alice has a good lawyer, Bob might lose in court. However, the signature scheme's failure to resist [attacks on S-DEO] at the very least muddies the water. The cryptographic weakness turns a simple court case into a more difficult one.

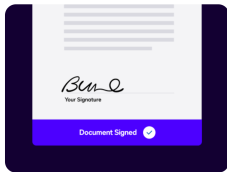
Moreover, Alice might lose in court for the simple reason that judges and jurors would be inclined to regard a signature scheme as faulty if it allows for different parties to have identical signatures on a document. That is, Bob's lawyer might not have to reconstruct a plausible scenario in which Bob did not really sign the document; it might be enough for him to impugn the credibility of the entire signature scheme.

“THE EQUIVALENT OF A WRITTEN SIGNATURE” [DH76]

Although now firmly established in the security literature, non-repudiation never fully gained traction as a legal concept, as it seemed to preempt the very process by which the qualities of a given form of evidence are gradually established through the adversarial process. — [Bla12, p.11]

“THE EQUIVALENT OF A WRITTEN SIGNATURE” [DH76]

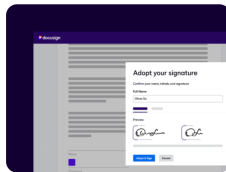
Docusign supports all levels of signatures defined by eIDAS



Electronic Signature

Docusign eSignature is trusted by hundreds of millions of users world-wide and meets electronic signature regulations, including eIDAS.

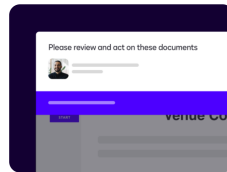
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Advanced Electronic Signature

Add Docusign ID Verification to your electronic signature process to validate signer identity and support AES. This is needed in use cases that require stricter identity assurance.

[Learn More >](#)



Qualified Electronic Signatures

A QES is the legal equivalent of a written signature and has special legal status in the EU. Meet EU and UK QES standards by combining eSignature with our ID Verification for EU Qualified offering or partner solutions.

[Learn More >](#)

DENIABILITY

- In some applications, the opposite of non-repudiation may be desirable: deniability.
- Deniability is the guarantee that I can deny having sent a certain message after the fact.
- It may be desirable in a one-to-one chat to prevent Bob from taking Alice' messages and present these to Charley to convince him of the mean things Alice has said about him.
- Not being able to do that is closer to a face to face conversation than being able to.
- Like with “non-repudiation”, the law may not care much about “deniability” at the cryptographic level

Reading Recommendation

Matthew Green. *Ok Google: please publish your DKIM secret keys*. 2020

<https://blog.cryptographyengineering.com/2020/11/16/>

EXAMPLE: AT-COMPROMISE SECURITY

We talk about abductions and torture.

Reference

Martin R. Albrecht, Simone Colombo, Benjamin Dowling, and Rikke Bjerg Jensen.
At-Compromise Security: The Case for Alert Blindness. Cryptology ePrint Archive, Paper 2026/252. 2026. URL: <https://eprint.iacr.org/2026/252>

ESTABLISHING SOCIAL FOUNDATIONS

1. Through **ethnographic findings** we analyse **social relations**,
2. to **discover security concerns** and study them with **cryptography**.

CONTEXT: GEN-Z PROTESTS IN KENYA



- Unprecedented **large-scale protests** in 2024, sparked by the 2024 **Finance Bill**.
- **25 June 2024**: protesters breached the fence around Kenya's Parliament, some entered Parliament and the Senate. State forces killed and injured protesters.
- The 2024 Finance Bill was **withdrawn on 26 June**; smaller-scale protests until 8 August.

Abductions by William Ruto's repressive forces emerged as a new threat during these protests.

- June-December 2024: 82 abductions [Ken24].
- 25 June 2024: coordinated abductions of at least 50 (known) protesters.
- Additional **15** abductions linked to the one-year anniversary protests [Ken25].

The Current Situation

The whereabouts of several abducted people remain unknown.

Ethnography is a methodology that comprises **data collection methods** and **analytical steps**.

- [ACD]26] introduces **ethnography to cryptography**.
- It is through the application of this technique that they arrived at their new security notion.

Long immersion in a social movement in Kenya that opened access to a wide network of activists.



Active participant observation

- 11 protests in Nairobi.
- Activist meetings, planning sessions, strategic discussions, community gatherings.
- Day-to-day activities in informal settlements (community outreach, hospital and prison visits, group meetings)

Ethnographic interviews in the field

- 77 protesters (organisers, frontliners, first-timers, seasoned; 8 abducted).
- Loosely structured around a topic guide.
- Duration: 43 minutes to over two hours.

- Analysis across ethnography and cryptography: during, between, after fieldwork.
- **Inductive reasoning**: theory developed from empirical data, not from hypotheses.
- Did not test an intuited security notion, but analysed the data to **discover** it.
- Initial analysis by fieldworker, followed by collective analysis and formalisation with cryptographers and fieldworker.

THE CONTOURS OF ABDUCTIONS: INTERROGATION

“[I]t’s psychological torture, because they [interrogators] tell you: ‘Just remember, you are not going back [...] so it depends on how you answer us [...]. Who is financing you? What are you planning?’ [...] They were using a tool. I don’t know whether it was an iron or wood, hitting me on the arm. So, I still have injuries [rolling up their sleeve to show the scars].” — (P72, abducted individual, Nairobi)

RAISING THE ALARM

During the Gen-Z protests, the practice of 'raising the alarm' emerged:

"[...] I wrote to him a message. Very brief message: 'Hey, I'm being abducted. I don't know where I am' [...] After I had texted him, he took the screenshot [of the message] and posted it on X. So that is how it spread [...] funnily enough, after like 30 minutes or so, as they [the abductors] were driving me around, the car stopped again. [...] They opened the trunk, and then they started beating me up. And then they asked me: 'Where is that phone?' So they knew that I had another phone because my friend shared my screenshot [...] I think that is the only thing that saved me. Because if it wasn't for that message, these guys had another motive. Because nobody was aware of my abduction, apart from that one message. So they could have killed me and taken me somewhere, nobody could have known."
— (P51, abducted individual, Nairobi)

[./lecture-social-foundations-posters-free.png] Posters calling for the release of abducted protesters.

- Once the alarm had been raised, shouting could happen across platforms.
- A network of connected and known activists amplified the message.
- This created a feedback loop: the abductors eventually heard the message, which put pressure on them to release the abducted person.

AT-COMPROMISE ALERT BLINDNESS

When they are taken, abducted people can alert their network without the adversary noticing.

SIMPLIFIED SYNTAX

Client

Server

.....end of setup phase and start of alert phase

SIMPLIFIED SYNTAX

Client

Server

$(ds, cm) \leftarrow \text{ClientSetup}(\cdot, k, ps)$

..... end of setup phase and start of alert phase

SIMPLIFIED SYNTAX

Client

Server

$(ds, cm) \leftarrow \text{ClientSetup}(\cdot, k, ps)$

cm



..... end of setup phase and start of alert phase

SIMPLIFIED SYNTAX

Client

Server

$(ds, cm) \leftarrow \text{ClientSetup}(\cdot, k, ps)$

cm



$(\cdot, ss, \cdot) \leftarrow \text{ServerSetup}(\cdot, cm)$

..... end of setup phase and start of alert phase

SIMPLIFIED SYNTAX

Client

Server

$(ds, cm) \leftarrow \text{ClientSetup}(\cdot, k, ps)$

cm

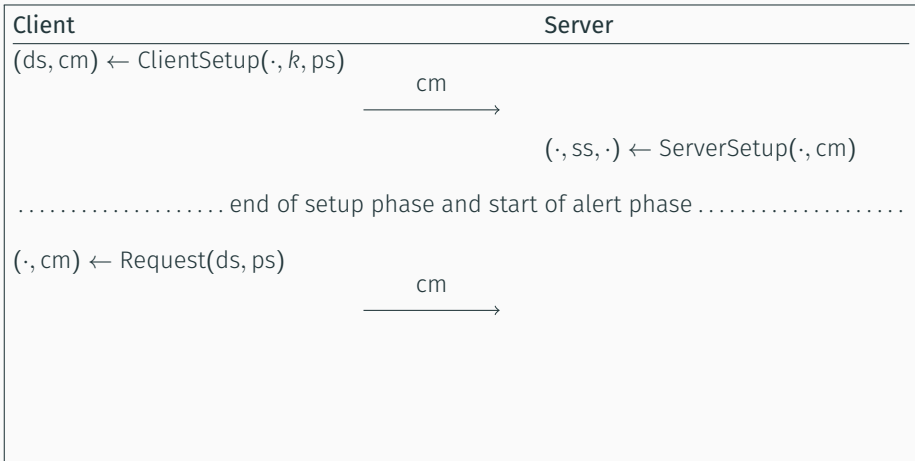


$(\cdot, ss, \cdot) \leftarrow \text{ServerSetup}(\cdot, cm)$

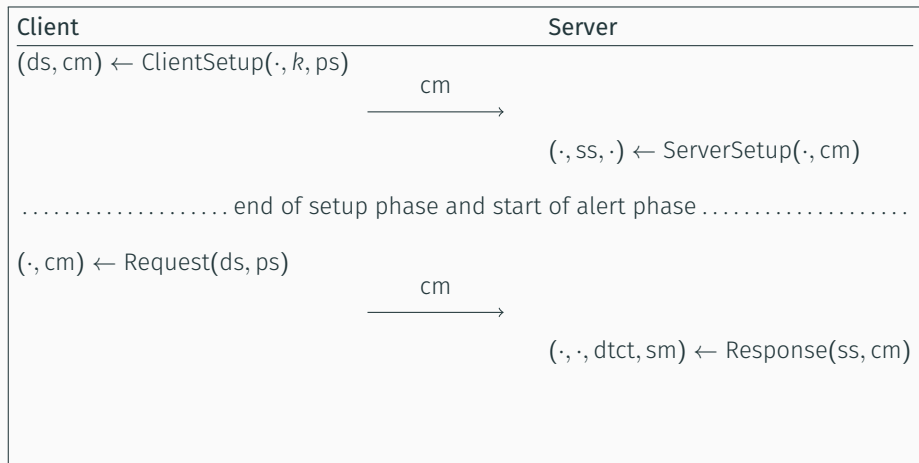
..... end of setup phase and start of alert phase

$(\cdot, cm) \leftarrow \text{Request}(ds, ps)$

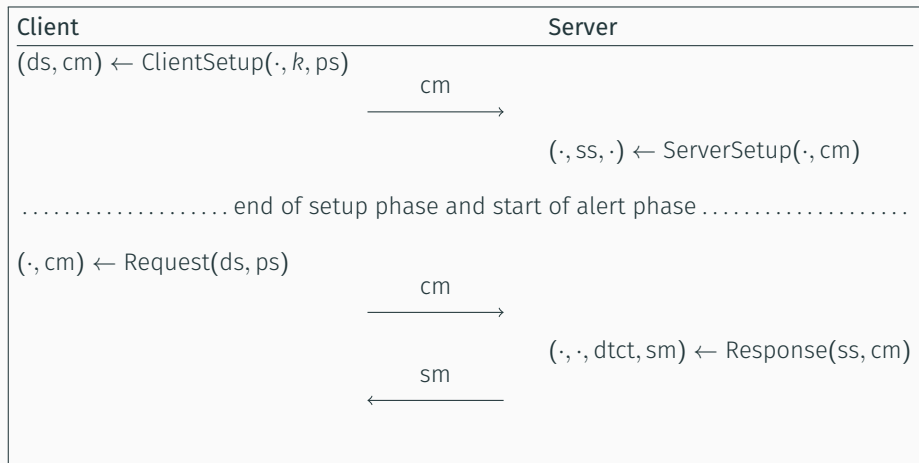
SIMPLIFIED SYNTAX



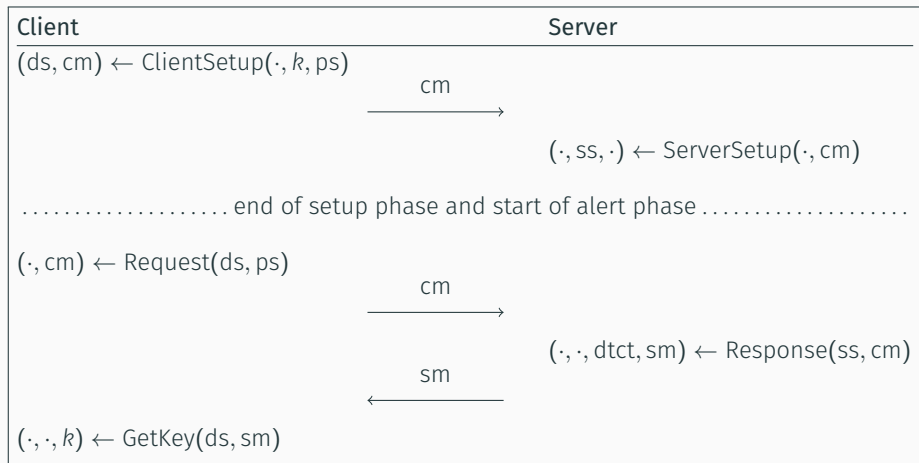
SIMPLIFIED SYNTAX



SIMPLIFIED SYNTAX



SIMPLIFIED SYNTAX



SECURITY NOTION: KIND-AtC-AB

Game KIND-AtC-AB

$ps_1 \leftarrow \mathcal{S}; ps_0 \leftarrow \mathcal{S} \setminus \{ps_1\}; k_1, k_0 \leftarrow \mathcal{K}$
 $(ds, ss) \leftarrow \text{GeneralSetup}(1^\lambda, k_1, ps_1)$
 $\text{win}_{cm}, \text{win}_{sm} \leftarrow \text{false}; d \leftarrow \{0, 1\}; b \leftarrow \{0, 1\}$
 $\sigma \leftarrow \mathcal{A}^{\text{REQUEST}, \text{RESPONSE}, \text{GETKEY}, \text{CLEAR}}$
if $\text{win}_{cm} \vee \text{win}_{sm}$ **then return** 1
 $ds \leftarrow \text{Clear}(ds)$
 $\text{dtct}, \text{recv} \leftarrow \text{false}$
 $(b', d') \leftarrow \mathcal{A}^{\text{DETECT}}(\sigma, ds, ps_b)$
if $d = d' \wedge \neg \text{recv}$ **then return** 1
if $b = 0 \wedge \text{recv} \wedge \neg \text{dtct}$ **then return** 1
if $b = b'$ **then return** 1
return 0

Oracle REQUEST()

$(ds, cm) \leftarrow \text{Request}(ds, ps_1)$
 $\mathcal{C} \leftarrow \mathcal{C} \cup \{cm\}$
return cm

Oracle RESPONSE(cm)

$(\text{acc}, ss, \cdot, sm) \leftarrow \text{Response}(ss, cm)$
if $\text{acc} \wedge cm \notin \mathcal{C}$ **then** $\text{win}_{cm} \leftarrow \text{true}$
 $\mathcal{Z} \leftarrow \mathcal{Z} \cup \{sm\}$
return sm

Oracle GETKEY(sm)

$(\text{acc}, ds, \cdot) \leftarrow \text{GetKey}(ds, sm)$
if $\text{acc} \wedge sm \notin \mathcal{Z}$ **then** $\text{win}_{sm} \leftarrow \text{true}$

Oracle DETECT(cm)

$(\text{acc}, ss, \text{dtct}', sm) \leftarrow \text{Response}(ss, cm)$
if $\neg \text{acc}$ **then return** $\perp$
 $\text{recv} \leftarrow \text{true}$
 $\text{dtct} \leftarrow \text{dtct} \vee \text{dtct}'$
return sm

Oracle CLEAR()

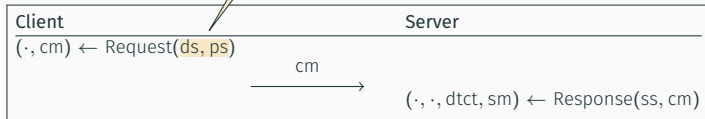
$ds \leftarrow \text{Clear}(ds)$

REALISM: PERSONAL SECRETS

Game KIND-AtC-AB

```
 $ps_1 \leftarrow \mathcal{S}; ps_0 \leftarrow \mathcal{S} \setminus \{ps_1\}; k_1, k_0 \leftarrow \mathcal{K}$   
[...]  
 $(b', d') \leftarrow \mathcal{A}^{\text{DETECT}, \text{ENC}}(\sigma, ds, ps_b)$   
if  $d = d' \wedge \neg \text{recv}$  then return 1  
if  $b = 0 \wedge \text{recv} \wedge \neg \text{dtct}$  then return 1  
if  $b = b'$  then return 1  
return 0
```

Device state ds and personal secret ps are **different** things.

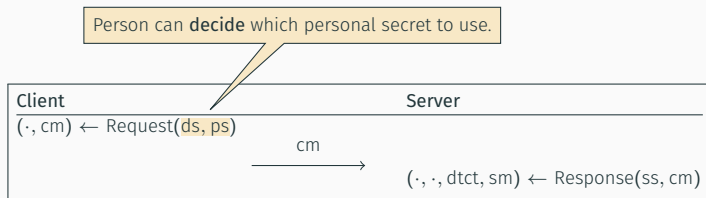


- Personal secrets capture the **distinction** of compromising people or their devices.

REALISM: PERSONAL SECRETS

Game KIND-AtC-AB

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 $ps_1 \leftarrow \mathcal{S}; ps_0 \leftarrow \mathcal{S} \setminus \{ps_1\}; k_1, k_0 \leftarrow \mathcal{K}$   
[...]  
 $(b', d') \leftarrow \mathcal{A}^{\text{DETECT}, \text{ENC}}(\sigma, ds, ps_b)$   
if  $d = d' \wedge \neg \text{recv}$  then return 1  
if  $b = 0 \wedge \text{recv} \wedge \neg \text{dtct}$  then return 1  
if  $b = b'$  then return 1  
return 0
```



- Personal secrets capture the **distinction of compromising people or their devices**.
- Multiple personal secrets model that the abducted person knows they have been abducted: **they can decide which secret to use**.

Game KIND-AtC-AB

[...]

$(b', d') \leftarrow \mathcal{A}^{\text{DETECT}, \text{ENC}}(\sigma, ds, ps_b)$

if $d = d' \wedge \neg \text{recv}$ **then return** 1

[...]

return 0

Oracle DETECT(cm)

$(\text{acc}, ss, \text{dtct}', sm) \leftarrow \text{Response}(ss, cm)$

if $\neg \text{acc}$ **then return** $\perp$

$\text{recv} \leftarrow \text{true}$

AEAD rejection

$\text{dtct} \leftarrow \text{dtct} \vee \text{dtct}'$

return sm

Game KIND-AtC-AB

[...]

$(b', d') \leftarrow \mathcal{A}^{\text{DETECT}, \text{ENC}}(\sigma, ds, ps_b)$

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$(\text{acc}, ss, \text{dtct}', sm) \leftarrow \text{Response}(ss, cm)$

if $\neg \text{acc}$ **then return** $\perp$

$\text{recv} \leftarrow \text{true}$

$\text{dtct} \leftarrow \text{dtct} \vee \text{dtct}'$

return sm

- “They wanted to know who was funding these protest.” (P49)

REALISM: CONFIDENTIALITY

Adversary wins if they recover KIND bit d **without** sending a valid client message.

Game KIND-AtC-AB

[...]

$(b', d') \leftarrow \mathcal{A}^{\text{DETECT, ENC}}(\sigma, ds, ps_b)$

if $d = d' \wedge \neg \text{recv}$ then return 1

[...]

return 0

Oracle ~~DETECT~~(cm)

$(\text{acc}, ss, \text{dtct}', sm) \leftarrow \text{Response}(ss, cm)$

if $\neg \text{acc}$ then return $\perp$

recv $\leftarrow \text{true}$

$\text{dtct} \leftarrow \text{dtct} \vee \text{dtct}'$

return sm

- “They wanted to know who was funding these protest.” (P49)
- We exploit the abductors’ goal to access information: breaking KIND requires **at least** one client’s message to reach the server (**recv $\leftarrow \text{true}$**).

Game KIND-AtC-AB

$ps_1 \leftarrow \mathcal{S}; ps_0 \leftarrow \mathcal{S} \setminus \{ps_1\}; k \leftarrow \mathcal{K}$
[...]
 $(b', d') \leftarrow \mathcal{A}^{\text{DETECT}, \text{ENC}}(\sigma, ds, ps_b)$
if $d = d' \wedge \neg \text{recv}$ then return 1
if $b = 0 \wedge \text{recv} \wedge \neg \text{dtct}$ then return 1
[...]

Oracle DETECT(cm)

$(\text{acc}, ss, \text{dtct}', sm) \leftarrow \text{Response}(ss, cm)$
if $\neg \text{acc}$ then return $\perp$
 $\text{recv} \leftarrow \text{true}$
 $\text{dtct} \leftarrow \text{dtct} \vee \text{dtct}'$
return sm

Game KIND-AtC-AB

```
ps1 ←  $\mathcal{S}$ ; ps0 ←  $\mathcal{S} \setminus \{ps_1\}$ ; k ←  $\mathcal{K}$   
[...]  
(b', d') ←  $\mathcal{A}^{\text{DETECT}, \text{ENC}}(\sigma, ds, ps_b)$   
if d = d' ∧ ¬recv then return 1  
if b = 0 ∧ recv ∧ ¬dtct then return 1  
[...]
```

Oracle DETECT(cm)

```
(acc, ss, dtct', sm) ← Response(ss, cm)  
if ¬acc then return ⊥  
recv ← true  
dtct ← dtct ∨ dtct'  
return sm
```

Make disappearance publicly known: “*I think that is the only thing that saved me.*” (P51)

REALISM: ALERT

Game KIND-AtC-AB

$ps_1 \leftarrow \mathcal{S}; ps_0 \leftarrow \mathcal{S} \setminus \{ps_1\}; k \leftarrow \mathcal{K}$
[...]
 $(b', d') \leftarrow \mathcal{A}^{\text{DETECT, ENC}}(\sigma, ds, ps_b)$
if $d = d' \wedge \neg \text{recv}$ then return 1
if $b = 0 \wedge \text{recv} \wedge \neg \text{dtct}$ then return 1
[...]

Oracle DETECT(cm)

$(\text{acc}, ss, \text{dtct}', sm) \leftarrow \text{Response}(ss, cm)$
if $\neg \text{acc}$ then return $\perp$
 $\text{recv} \leftarrow \text{true}$
 $\text{dtct} \leftarrow \text{dtct} \vee \text{dtct}'$
return sm

Adversary wins if given **any random** $ps_0 \leftarrow \mathcal{S}$ different from the setup personal secret ps_1 they manage to **not** trigger an alarm.

Make disappearance publicly known: “*I think that is the only thing that saved me.*” (P51)

REALISM: BLINDNESS

Game KIND-AtC-AB

$b \leftarrow \{0, 1\}$

[...]

$(b', d') \leftarrow \mathcal{A}^{\text{DETECT, ENC}}(\sigma, ds, ps_b)$

[...]

if $b = b'$ then return 1

return 0

Client

Server

$(\cdot, \cdot, dtct, sm) \leftarrow \text{Response}(ss, cm)$

sm



$(\cdot, \cdot, k) \leftarrow \text{GetKey}(ds, sm)$

- Non-cooperation: “If I do anything now to upset these guys, I’m dead.” (P60)

REALISM: BLINDNESS

Game KIND-AtC-AB

$b \leftarrow \{0, 1\}$

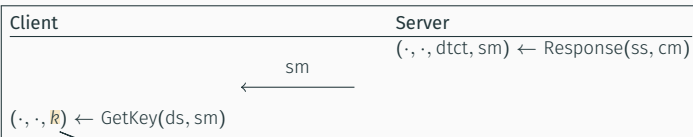
[...]

$(b', d') \leftarrow \mathcal{A}^{\text{DETECT, ENC}}(\sigma, ds, ps_b)$

[...]

if $b = b'$ then return 1

return 0



The protocol **always** reconstructs the key.

- **Non-cooperation:** “If I do anything now to upset these guys, I’m dead.” (P60)
- The protocol **always** reconstructs the key: abducted persons would cooperate regardless.

REALISM: BLINDNESS

Game KIND-AtC-AB

$b \leftarrow \{0, 1\}$

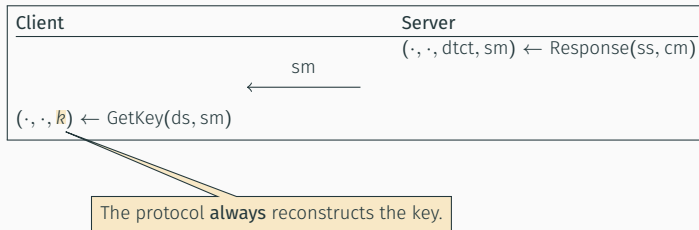
[...]

$(b', d') \leftarrow \mathcal{A}^{\text{DETECT, ENC}}(\sigma, ds, ps_b)$

[...]

if $b = b'$ then return 1

return 0



- **Non-cooperation:** “If I do anything now to upset these guys, I’m dead.” (P60)
- The protocol **always** reconstructs the key: abducted persons would cooperate regardless.
- We **give up confidentiality for alert:** a goal **only ethnography** could suggest.

REALISM: BLINDNESS

Game KIND-AtC-AB

$b \leftarrow \{0, 1\}$

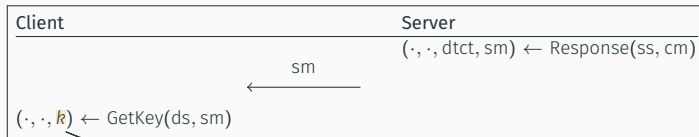
[...]

$(b', d') \leftarrow \mathcal{A}^{\text{DETECT, ENC}}(\sigma, ds, ps_b)$

[...]

if $b = b'$ then return 1

return 0



- **Non-cooperation:** “If I do anything now to upset these guys, I’m dead.” (P60)
- The protocol **always** reconstructs the key: abducted persons would cooperate regardless.
- We **give up confidentiality for alert:** a goal **only ethnography** could suggest.

This notion is achievable: they give a provably secure construction and they implement it.

FIN

“WHILE THE IMMEDIATE OBJECTS OF CRYPTOGRAPHY ARE NOT
SOCIAL RELATIONS, IT PRESUMES AND MODELS THEM.” — [https://
social-foundations-of-cryptography.gitlab.io/about](https://social-foundations-of-cryptography.gitlab.io/about)

- [ACD]26] Martin R. Albrecht, Simone Colombo, Benjamin Dowling, and Rikke Bjerg Jensen. **At-Compromise Security: The Case for Alert Blindness**. Cryptology ePrint Archive, Paper 2026/252. 2026. URL: <https://eprint.iacr.org/2026/252>.
- [Bla12] Jean-François Blanchette. **Burdens of proof: Cryptographic culture and evidence law in the age of electronic documents**. MIT Press, 2012.
- [CDFFJ21] Cas Cremers, Samed DüzlÜ, Rune Fiedler, Marc Fischlin, and Christian Janson. **BUFFing signature schemes beyond unforgeability and the case of post-quantum signatures**. In: *2021 IEEE Symposium on Security and Privacy*. IEEE Computer Society Press, May 2021, pp. 1696–1714. DOI: 10.1109/SP40001.2021.00093.

- [DH76] Whitfield Diffie and Martin E. Hellman. **New Directions in Cryptography**. In: *IEEE Transactions on Information Theory* 22.6 (1976), pp. 644–654. DOI: 10.1109/TIT.1976.1055638.
- [Gee98] Clifford Geertz. **Deep Hanging Out**. In: *The New York Review of Books* 45.16 (1998), pp. 69–72.
- [GGSW13] Sanjam Garg, Craig Gentry, Amit Sahai, and Brent Waters. **Witness encryption and its applications**. In: *45th ACM STOC*. Ed. by Dan Boneh, Tim Roughgarden, and Joan Feigenbaum. ACM Press, June 2013, pp. 467–476. DOI: 10.1145/2488608.2488667.
- [Gol04] Oded Goldreich. **Foundations of Cryptography: Basic Applications**. Vol. 2. Cambridge, UK: Cambridge University Press, 2004. ISBN: 978-0511721656. DOI: 10.1017/CB09780511721656.

REFERENCES III

- [Gol97] Oded Goldreich. **On the Foundations of Modern Cryptography (Invited Lecture)**. In: *CRYPTO'97*. Ed. by Burton S. Kaliski Jr. Vol. 1294. LNCS. Springer, Berlin, Heidelberg, Aug. 1997, pp. 46–74. DOI: 10.1007/BFb0052227.
- [Ken24] Kenya National Commission on Human Rights. **Statement on the Recent Surge of Abductions/Enforced Disappearances in Kenya**. Accessed: 2025-07-01. 2024. URL: <https://www.knchr.org/Articles/ArtMID/2432/ArticleID/1213/Statement-on-the-Recent-Surge-of-AbductionsEnforced-Disappearances-in-Kenya>.
- [Ken25] Kenya National Commission on Human Rights. **KNCHR Update on the 25th June 2025 Demonstrations**. Accessed: 2025-07-01. 2025. URL: <https://www.knchr.org/Articles/ArtMID/2432/ArticleID/1230/KNCHR-UPDATE-ON-THE-25TH-JUNE-2025-DEMONSTRATIONS>.

REFERENCES IV

- [KM11] Neal Koblitz and Alfred Menezes. **Another Look at Security Definitions**. Cryptology ePrint Archive, Report 2011/343. 2011. URL: <https://eprint.iacr.org/2011/343>.
- [KM13] Neal Koblitz and Alfred Menezes. **Another look at security definitions**. In: *Adv. Math. Commun.* 7.1 (2013), pp. 1–38. DOI: [10.3934/AMC.2013.7.1](https://doi.org/10.3934/AMC.2013.7.1). URL: <https://doi.org/10.3934/amc.2013.7.1>.
- [RSA78] Ronald L. Rivest, Adi Shamir, and Leonard M. Adleman. **A Method for Obtaining Digital Signatures and Public-Key Cryptosystems**. In: *Communications of the Association for Computing Machinery* 21.2 (Feb. 1978), pp. 120–126. DOI: [10.1145/359340.359342](https://doi.org/10.1145/359340.359342).
- [RST01] Ronald L. Rivest, Adi Shamir, and Yael Tauman. **How to Leak a Secret**. In: *ASIACRYPT 2001*. Ed. by Colin Boyd. Vol. 2248. LNCS. Springer, Berlin, Heidelberg, Dec. 2001, pp. 552–565. DOI: [10.1007/3-540-45682-1_32](https://doi.org/10.1007/3-540-45682-1_32).

- [Sch90] Claus-Peter Schnorr. **Efficient Identification and Signatures for Smart Cards**. In: *CRYPTO'89*. Ed. by Gilles Brassard. Vol. 435. LNCS. Springer, New York, Aug. 1990, pp. 239–252. DOI: 10.1007/0-387-34805-0_22.